A COMPATIBLE MAP OF STRUCTURE SHEAVES OF THICKENINGS COMES FROM A UNIQUE MORPHISM OF THICKENINGS OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author leaves two verifications to the reader at the end of the proof: that f' o i_X = i_Y o f (obtained by a second use of the fully faithful theorem) and that f' is a morphism over B (proof lines 1642-1645). Both are routine checks of the construction that is carried out inside the span.

Allowed standard prerequisites (editorial).

- Morphisms between thickenings correspond to morphisms of the associated ringed topoi. Properties of Spaces, Theorem labelled spaces-properties-theorem-fully-faithful; invoked at lines 1624-1626 and again at 1643-1644. Reduces the existence and uniqueness of f' to producing a morphism of locally ringed topoi from the given algebra map alpha.
- Definition of a morphism of locally ringed topoi. Modules on Sites, Definition labelled sites-modules-definition-morphism-locally-ringed-topoi; invoked at lines 1634-1636. Fixes exactly what has to be verified for (f_{spaces,etale}, alpha) and uses the local nilpotence of the kernels of i_X^sharp and i_Y^sharp.
- The underlying morphism is a morphism of locally ringed topoi. Properties of Spaces, Lemma labelled spaces-properties-lemma-morphism-locally-ringed; invoked at lines 1637-1639. Supplies the locally-ringed property for (f, f^sharp) against which alpha is compared.

Source reference. Original lemma 04HA: sources/stacks-chow/tags.txt, source lines 5861–5861. Navigation pointer for \ref{sites-modules-definition-morphism-locally-ringed-topoi} used inside the counted span (line 1636); the argument that consumes it is retained in the span.

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Source reference. Original lemma 04KJ: sources/stacks-chow/tags.txt, source lines 5975–5975. Navigation pointer for \ref{spaces-properties-lemma-morphism-locally-ringed} used inside the counted span (line 1639); the argument that consumes it is retained in the span.

Source reference. Original lemma 04KL: sources/stacks-chow/tags.txt, source lines 5977–5977. Navigation pointer for \ref{spaces-properties-theorem-fully-faithful} used inside the counted span (line 1625); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $X \subset X'$ and $Y \subset Y'$ be thickenings of algebraic spaces over B. Let $f : X \to Y$ be a morphism of algebraic spaces over B. Given any map of $\mathcal{O}_B$-algebras*

$$\alpha : f^{-1}_{spaces,étale}\mathcal{O}_{Y'} \to \mathcal{O}_{X'}$$

such that

$$\begin{array}{ccc} f^{-1}_{spaces,étale}\mathcal{O}_Y & \xrightarrow{f^\sharp} & \mathcal{O}_X \\ {\scriptstyle i_Y^\sharp}\uparrow & & \uparrow{\scriptstyle i_X^\sharp} \\ f^{-1}_{spaces,étale}\mathcal{O}_{Y'} & \xrightarrow{\alpha} & \mathcal{O}_{X'} \end{array}$$

commutes, there exists a unique morphism of (f, f') of thickenings over B such that $\alpha = (f')^\sharp$.

Proof. To find f', by Properties of Spaces, Theorem 04KL, all we have to do is show that the morphism of ringed topoi

$$(f_{spaces,étale}, \alpha) : (Sh(X_{spaces,étale}), \mathcal{O}_{X'}) \longrightarrow (Sh(Y_{spaces,étale}), \mathcal{O}_{Y'})$$

is a morphism of locally ringed topoi. This follows directly from the definition of morphisms of locally ringed topoi (Modules on Sites, Definition 04HA), the fact that $(f, f^\sharp)$ is a morphism of locally ringed topoi (Properties of Spaces, Lemma 04KJ), that α fits into the given commutative diagram, and the fact that the kernels of $i_X^\sharp$ and $i_Y^\sharp$ are locally nilpotent. Finally, the fact that $f' \circ i_X = i_Y \circ f$ follows from the commutativity of the diagram and another application of Properties of Spaces, Theorem 04KL. We omit the verification that f' is a morphism over B. □